Assignment: Contextual Visual-RAG Pipeline

Objective: Build a RAG pipeline that ingests and queries over PDFs to retrieve both **textual answers** and the **relevant images**. The responses should contain citations too if possible that are returned in the response as clickable links to the respective PDF pages. (Citations should ideally link to the specific page in the PDF viewer).

Tools: You can use tools and libraries of your choice depending on requirement and ongoing technologies in the area.

Task Overview:

- **Ingestion & Mapping:** Parse a multi-page PDF to extract text and images. Create and store the vector embeddings in a suitable vector database.

- **Retrieval Logic:** Implement semantic search to find relevant text chunks. If a retrieved chunk is linked to an image in the metadata, the system must identify it.

  (*Need not sematically search for images for the query, just need to associate relevant images with the text response if any*).

  Clearly document your strategy for maintaining the spatial or semantic relationship between and image and it's surrounding text.

- **Hallucination Control:** Restrict the LLM to only use the retrieved document context for its response.

- **API & UI:**

  * **Backend:** Create relevant API endpoints required for PDF ingestion and querying. Any framework can be used but fastapi is preferred.

  ***Frontend:** Build a simple interface (using any tools or frameworks deemed fit) where a user can upload a PDF and ask questions, seeing both the text response and the corresponding image side-by-side. The user should also be able to delete existing PDFs (including removal of the file chunks from the vector database along with its image context).

  The UI should display the PDF files that have been ingested into the pipeline again with clickable links.

  The system should be able to handle asynchronous PDF uploads and queries simultaneously.

Deliverables:

GitHub Repo: Clean code with a requirements.txt and a README.md explaining your text-to-image mapping logic and instructions on deploying and using the application (docker setup preferred).